

Worksheet 5

Exercise 1: Neighborhood of schools

The data frame `SchoolingReturns` in the file `schoolingreturns.rda` contains a part of the observational data from a U.S. national longitudinal survey of young men conducted in 1976. The aim of this exercise is to estimate the causal effect of education (`education` = education in 1976 in years) on salary (`wage` = raw wages in 1976 in cents per hour). We assume that there is an unobserved confounding factor, such as motivation, that has a causal effect on education and salary. In addition, there are two more variables: `experience` (years of labor market experience) and `nearcollege` (did the individual grow up near a college).

- Draw a DAG that visualizes the assumed causal relationships between the 4 variables given in the data.frame `SchoolingReturns` and the assumed confounder. Give reasons why the neighborhood to a college might be an appropriate instrument for education.
- Check if the variable `nearcollege` can be used as instrument for education. Use an appropriate visualization and test, to confirm that the variable `nearcollege` and `education` are associated.
- Fit a naive regression model taking the log of `wage` as outcome and `education` as explanatory variable. Determine the estimated effect of education on log-wage and explain why this effect might not correspond to the causal effect of education on log-wage.
- Use the function `ivreg` to fit an IV-regression model adding `nearcollege` as instrument for education. Compare and discuss the estimated effect of education on wage.
- Reproduce the result from d) with a 2-step-least-square (SLS).
- Now add `experience` as additional explanatory variable.
Note: Since `experience` has no instrument, you need to use `experience` itself as instrument in the `ivreg` function.
- Repeat the IV-regression from e) by doing 2SLS manually.

Exercise 2: Cigarette Demand in the United States

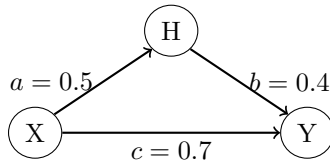
The `CigaretteDemand` dataset in the R-Package `ivreg` contains information on cigarette demand in the 48 contiguous U.S. states for the years 1985 and 1995. It includes variables such as per capita consumption, price, income, and taxes to analyze the economic determinants of cigarette consumption.

- Fit a linear regression with the response `log(packs)` and the two explanatory variables `log(rprice)` and `log(rincome)`.
- Fit an IV regression model (using logarithmic transformations as in part a), with `salestax` used as an instrument for the price. Discuss the estimated effect.

Exercise 3: Exam Score

A school has set up a supervised remedy program, to give students the opportunity to exercise in addition to their homework. The causal graph for the time spent in the remedy program (X), the time spent with homework (H) and the score achieved in the exam (Y) is depicted below. You can assume a linear SCM (for coefficients see the causal graph) and standard normal distributed noise terms.

(Reference: Judea Pearl et al, *Causal Inference in Statistics - A Primer*)



- What is the distribution of the exam score?
- Let us consider a student named Joe, who spent 0.5 hours in the supervised remedy program (i.e. $x = 0.5$) and spent double time for homework (i.e. $h = 1$). In the exam Joe achieved a score of $y = 1.5$. What would Joe's score have been had he doubled his study time of homework?
- Let us consider another student named Emily, who spend the same time as Joe in the supervised remedy program (i.e. $x = 0.5$). However, she spent double time as Joe studying (i.e. $h = 2$) and achieved a score of $y = 2.2$ in the exam. What would Emily's score have been had she studied same time as Joe?

Exercise 4: Ice Cream

Ice cream sales are influenced by various factors. One major factor is temperature. People tend to buy more ice cream in summer and less in winter. Advertisement can also influence how much ice cream is sold. Note that temperature is a driver for advertising ice cream, too.

- Draw a DAG that visualizes the described situation.
- Assume a linear SCM with normal distributed noise and estimate all causal effects from the observed data in the file `ice_cream.rda`.
- Have a look at the observation with the highest daily temperature.
 - How many ice creams were sold on that day?
 - How many ice-creams would have been sold on that particular day, if the temperature would have been only 30?
 - How many ice-creams would have been sold on that particular day, if there would have been no advertisement?
- Use linear regression to calculate a usual prediction for the two data point described as counterfactual in c). Explain the difference.